Udyam Saathi · Solution architecture

Project Udyam Digital for Accel Business — an MSME owner goes from first question to an active connection without a sales visit.

Request · Journey hand-off · Mocked for the prototype

01 Discover	02 Quote	03 Verify	04 Activate	05 Operate
AI advisor finds the right bundle	Price, adjust, accept	Paperless KYC	Install and go live	Score, route, review

USERS

MSME owner — Mobile browser · any Indian language · text or voice

Accel staff and partners

WEB APP — React PWA

- **Advisor chat** — Guided conversation in English or Hindi
- **Bundles & quote** — Three tiers to compare, edit and accept, priced with GST
- **Documents** — Consent, camera or DigiLocker, e-sign
- **Order tracker** — Live status to activation
- **Ops console** — Leads, routing, review queue

SERVICES — FastAPI · Python

- **Advisor** — LLM with tool calling; a guard blocks any price not from the engine
- tools →
- **Product engine** — Eligibility, pricing and tiers. Deterministic.
- ← price
- **Checkout** — Re-prices every change on the server; creates the order
- order →
- **Smart KYC** — OCR and ID checks, verdict with reasons
- verified →
- **Orders** — Fulfilment state machine and notifications
- **Console** — Propensity model routes each lead

EXTERNAL — APIs

- LIVE API · **Gemini** — Receives chat text only, never documents
- read / write
- MOCK · **Gov & identity** — GSTIN, Udyam, DigiLocker, e-Sign
- MOCK · **UPI & WhatsApp** — Autopay mandate, status messages

DATA — SQLite · files

- **Operational store** — SQLite documents: sessions, orders, KYC cases, consents, audit log
- **KYC vault** — Images encrypted at rest, deleted once a decision is made
- **Reference data** — Product catalogue, historical leads, registry files

TRUST

Consent first	Minimal data	Short retention	Accountable	Grounded AI
DPDP consent ledger, revocable	No IDs in chat; masked on screen	Encrypted, auto-deleted images	Audit log of consent, KYC and reviews	Prices come only from the engine

Prototype stack: React + TypeScript (Vite) · FastAPI · SQLite · Gemini API · Tesseract OCR

github.com/AgileSeagull/UdyamSaathi